Fatim Majumder

Machine Learning Research Engineer

New York, NY | fatim.majumder@emory.edu | github.com/fatimmajumder | linkedin.com/in/fatim-majumder

EDUCATION

Emory University - B.S. in Computer Science and Mathematics, Expected May 2026
- GPA: 3.96 / 4.00
- Coursework: algorithms, systems programming, machine learning, optimization, numerical analysis, mathematical statistics, real analysis, computer architecture

EXPERIENCE

Arthur AI - Machine Learning Research Engineering Intern, May 2025 to Aug 2025
- Built a production-grade LLM evaluation and model analysis platform across Python, PyTorch, Ray, FastAPI, PostgreSQL, Redis, Docker, Kubernetes, and AWS.
- Designed deterministic lineage across datasets, prompts, tokenizers, model artifacts, judges, and scoring logic; exact rerun reproducibility improved from 74.1% to 99.96%.
- Built scheduling, caching, retry semantics, and run-diff tooling; median experiment turnaround improved by 84% while reducing wasted GPU-hours by 51%.

Fullstory - Software Engineering Intern, May 2024 to Aug 2024
- Built backend observability and validation systems for analytics pipelines processing 4B+ weekly events using Python, SQL, Kafka, Airflow, dbt, and Docker.
- Developed validators for schema drift, freshness lag, replay mismatches, null spikes, cardinality explosions, and anomaly bursts.
- Improved alert precision from 63% to 96% and reduced median detection time from 3.4 hours to 7 minutes.

Algory Capital / Emory University - Head of Research, Sep 2023 to Present
- Built the research stack for a 30+ member student-run investment organization across ingestion, factor modeling, walk-forward backtesting, portfolio construction, and diagnostics.
- Engineered a shared platform in Python, SQL, PostgreSQL, Streamlit, Docker, and GitHub Actions; strategy research throughput increased 6.8x and analyst ramp time fell from about 6 weeks to 8 days.

Georgia Tech / Emory University - Research Assistant, Sep 2022 to Dec 2023
- Built reproducible multimodal ML pipelines for clinical and imaging datasets; improved held-out AUROC from 0.71 to 0.88 after eliminating leakage and cohort contamination.

PROJECTS

RE-AMP: Generative Audio Robustness Evaluation - Full-stack benchmarking platform with asynchronous orchestration, artifact storage, reproducible replay, and comparative analysis workflows.

Traffic Collision Risk Modeling - Graph-learning pipeline over large-scale collision and road-network data with spatial, temporal, and structural features.

Robust Kalman Filtering for Vehicle Localization - Simulation framework with sensor dropout, burst noise, outliers, covariance misspecification, and asynchronous measurements.

Quant Research Lab - Public-facing factor research and walk-forward backtesting demo with dashboard-first analysis.
- Research archive includes diffusion models, derivative-aware Gaussian processes, stochastic control, mirror descent, momentum methods, and optimizer analysis.

SKILLS

- Languages: Python, SQL, TypeScript, JavaScript, Java, C++, C, Bash, MATLAB
- ML / data: PyTorch, PyTorch Geometric, scikit-learn, XGBoost, Ray, LLM evaluation, graph ML, multimodal modeling, retrieval pipelines
- Systems: FastAPI, PostgreSQL, MySQL, Redis, Docker, Kubernetes, Linux, CI/CD, Airflow, dbt, Kafka, ETL, observability, distributed execution, caching

HONORS AND LEADERSHIP

- QuestBridge National Match Scholar; Dean’s List; Undergraduate Research Distinction; HackDuke finalist
- Head of Research at Algory Capital; leadership and community work across rowing, robotics, and WGRE radio programming